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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/704,167	04/24/2024	Hans KRAEMER	70028US01	5544

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Haleon
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EXAMINER	
POON, DANA LEE	

ART UNIT	PAPER NUMBER
3723	

NOTIFICATION DATE	DELIVERY MODE
04/22/2026	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

uspatents@haleon.com

Office Action Summary	Application No. 18/704,167	Applicant(s) KRAEMER, Hans	
	Examiner DANA L POON	Art Unit 3723	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 24 April 2024.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 15-28 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 15-28 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☑ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 24 April 2024 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☑ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☑ All b)☐ Some** c)☐ None of the:

1.☑ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Specification

2. The amendment filed 24 April, 2024 is objected to under 35 U.S.C. 132(a) because it introduces new matter into the disclosure. 35 U.S.C. 132(a) states that no amendment shall introduce new matter into the disclosure of the invention. The added material which is not supported by the original disclosure is as follows: The incorporation by reference of the international patent application PCT/EP2022/080884 and of the European Patent application No. 21207029.6 is ineffective as it was added on the date of entry into the national phase, which is after the filing date of the instant application. The filing date of this national stage application is the filing date of associated PCT, in this case 04 November, 2022, see MPEP 1893.03(b). Therefore the specification amendment of 24 April 2024 to include the incorporation by reference is new matter, per MPEP 608.01(p).

Applicant is required to cancel the new matter in the reply to this Office Action.

Claim Objections

3. **Claims 15-17, 19, 22, 24-28** are objected to because of the following informalities:

- Claims 15, 22, and 24-27 recites “the deformable element” that should be “the at least one deformable element”.
- Claims 15 and 24 recites “the cleaning zone” that should be “the at least one cleaning zone”.
- Claim 15 recites “the first arm” that should be “the first flexible arm”.
- Claims 15-17, 19, 22, and 28 recites “the arms” that should be “the first and second flexible arms”.
- Claims 15-17, 19, and 28 recites “the bridges” that should be “the at least two bridges”.

Appropriate correction is required.

Claim Rejections - 35 USC § 112

4. The following is a quotation of 35 U.S.C. 112(b):
(b) CONCLUSION.—The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the inventor or a joint inventor regards as the invention.

The following is a quotation of 35 U.S.C. 112 (pre-AIA), second paragraph:
The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the applicant regards as his invention.

5. **Claims 15-28** are rejected under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph, as being indefinite for failing to particularly point out and distinctly claim the subject matter which the inventor or a joint inventor (or for applications subject to pre-AIA 35 U.S.C. 112, the applicant), regards as the invention.

- The term "relatively towards" and "relatively away" in **claim 15** is a relative term which renders the claim indefinite. The term "relatively" is not defined by the claim, the specification does not provide a standard for ascertaining the requisite degree, and one of ordinary skill in the art would not be reasonably apprised of the scope of the invention. For purpose of examination examiner interprets "relatively towards" and "relatively away" as "towards" and "away".
- The term "substantially flat" in **claim 15** is a relative term which renders the claim indefinite. The term "relatively" is not defined by the claim, the specification does not provide a standard for ascertaining the requisite degree, and one of ordinary skill in the art would not be reasonably apprised of the scope of the invention. For purpose of examination examiner interprets "substantially flat" as "flat".
- The term "about" in **claims 18 and 21-23** are a relative term which renders the claim indefinite. The term "about" is not defined by the claim, the specification does not provide a standard for ascertaining the requisite degree, and one of ordinary skill in the art would not be reasonably

apprised of the scope of the invention. For purpose of examination examiner interprets the word "about" as deleted.

- The term "around" in **claim 28** are a relative term which renders the claim indefinite. The term "around" is not defined by the claim, the specification does not provide a standard for ascertaining the requisite degree, and one of ordinary skill in the art would not be reasonably apprised of the scope of the invention. For purpose of examination examiner interprets the word "around" as deleted.
- Claim 21 recites "the longitudinal direction". There is insufficient antecedent basis for this limitation in the claim. For purpose of examination, Examiner interprets the limitation as "a longitudinal direction".
- A broad range or limitation together with a narrow range or limitation that falls within the broad range or limitation (in the same claim) may be considered indefinite if the resulting claim does not clearly set forth the metes and bounds of the patent protection desired. See MPEP § 2173.05(c). In the present instance, **claim 18** recites the broad recitation "50 to 66", and the claim also recites "54 to 62", "56 to 60", and "58" which is the narrower statement of the range/limitation. The claim(s) are considered indefinite because there is a question or doubt as to whether the feature introduced by such narrower language is (a) merely exemplary of the remainder of the claim, and therefore not required, or (b) a required feature of the claims. Further the claim recites "preferably" and it is unclear to the examiner if applicant intends for the claimed range to be incorporated or not. For purpose of examination, examiner interprets the limitation having a range of 50 to 66.
- Claim 22 recites "the distance". There is insufficient antecedent basis for this limitation in the claim. For purpose of examination, Examiner interprets the limitation as "a distance".

- A broad range or limitation together with a narrow range or limitation that falls within the broad range or limitation (in the same claim) may be considered indefinite if the resulting claim does not clearly set forth the metes and bounds of the patent protection desired. See MPEP § 2173.05(c). In the present instance, **claim 22** recites the broad recitation "3mm to 5mm", and the claim also recites "3.5 to 4.5mm" which is the narrower statement of the range/limitation. The claim(s) are considered indefinite because there is a question or doubt as to whether the feature introduced by such narrower language is (a) merely exemplary of the remainder of the claim, and therefore not required, or (b) a required feature of the claims. Further the claim recites "preferably" and it is unclear to the examiner if applicant intends for the claimed range to be incorporated or not. For purpose of examination, examiner interprets the limitation having a range of 3mm to 5mm.
- A broad range or limitation together with a narrow range or limitation that falls within the broad range or limitation (in the same claim) may be considered indefinite if the resulting claim does not clearly set forth the metes and bounds of the patent protection desired. See MPEP § 2173.05(c). In the present instance, **claim 23** recites the broad recitation "5 to 9mm", and the claim also recites "7 to 8mm" which is the narrower statement of the range/limitation. The claim(s) are considered indefinite because there is a question or doubt as to whether the feature introduced by such narrower language is (a) merely exemplary of the remainder of the claim, and therefore not required, or (b) a required feature of the claims. Further the claim recites "preferably" and it is unclear to the examiner if applicant intends for the claimed range to be incorporated or not. For purpose of examination, examiner interprets the limitation having a range of 5 to 9mm.
- A broad range or limitation together with a narrow range or limitation that falls within the broad range or limitation (in the same claim) may be considered indefinite if the resulting claim does

not clearly set forth the metes and bounds of the patent protection desired. See MPEP § 2173.05(c). In the present instance, **claim 24** recites the broad recitation "not longer than 150%", and the claim also recites "not longer than 130%" which is the narrower statement of the range/limitation. The claim(s) are considered indefinite because there is a question or doubt as to whether the feature introduced by such narrower language is (a) merely exemplary of the remainder of the claim, and therefore not required, or (b) a required feature of the claims. Further the claim recites "preferably" and it is unclear to the examiner if applicant intends for the claimed range to be incorporated or not. For purpose of examination, examiner interprets the limitation having a range of not longer than 150%.

- A broad range or limitation together with a narrow range or limitation that falls within the broad range or limitation (in the same claim) may be considered indefinite if the resulting claim does not clearly set forth the metes and bounds of the patent protection desired. See MPEP § 2173.05(c). In the present instance, **claim 25** recites the broad recitation "2-7 bridges", and the claim also recites "3-5 bridges" and "3 bridges" which is the narrower statement of the range/limitation. The claim(s) are considered indefinite because there is a question or doubt as to whether the feature introduced by such narrower language is (a) merely exemplary of the remainder of the claim, and therefore not required, or (b) a required feature of the claims. Further the claim recites "preferably" and it is unclear to the examiner if applicant intends for the claimed range to be incorporated or not. For purpose of examination, examiner interprets the limitation having a range of 2-7 bridges.
- A broad range or limitation together with a narrow range or limitation that falls within the broad range or limitation (in the same claim) may be considered indefinite if the resulting claim does not clearly set forth the metes and bounds of the patent protection desired. See MPEP § 2173.05(c). In the present instance, **claim 27** recites the broad recitation "500 to 800%", and the

claim also recites “600 to 750%” and “650 to 700%” which is the narrower statement of the range/limitation. The claim(s) are considered indefinite because there is a question or doubt as to whether the feature introduced by such narrower language is (a) merely exemplary of the remainder of the claim, and therefore not required, or (b) a required feature of the claims. Further the claim recites “preferably” and it is unclear to the examiner if applicant intends for the claimed range to be incorporated or not. For purpose of examination, examiner interprets the limitation having a range of 500 to 800%.

- A broad range or limitation together with a narrow range or limitation that falls within the broad range or limitation (in the same claim) may be considered indefinite if the resulting claim does not clearly set forth the metes and bounds of the patent protection desired. See MPEP § 2173.05(c). In the present instance, **claim 28** recites the broad recitation “95° to 110”, and the claim also recites “97° to 105°” which is the narrower statement of the range/limitation. The claim(s) are considered indefinite because there is a question or doubt as to whether the feature introduced by such narrower language is (a) merely exemplary of the remainder of the claim, and therefore not required, or (b) a required feature of the claims. Further the claim recites “preferably” and it is unclear to the examiner if applicant intends for the claimed range to be incorporated or not. For purpose of examination, examiner interprets the limitation having a range of 95° to 110°.
- Claims 16-17, 19-20, and 26 are rejected due to being dependent upon a rejected claim.

Claim Rejections - 35 USC § 102

6. The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

A person shall be entitled to a patent unless –

(a)(1) the claimed invention was patented, described in a printed publication, or in public use, on sale, or otherwise available to the public before the effective filing date of the claimed invention.

(a)(2) the claimed invention was described in a patent issued under section 151, or in an application for patent published or deemed published under section 122(b), in which the patent or application, as the case may be, names another inventor and was effectively filed before the effective filing date of the claimed invention.

7. **Claims 15-17, 19-20, 24-26, and 28** are rejected under 35 U.S.C. 102(a)(1) as being anticipated by Kraemer (2009/0313779).

- Regarding Claim 15, Kraemer teaches A toothbrush (Fig. 1) with a handle (Ref. 2, Fig. 1), at least one cleaning zone (Ref. 4, Fig. 1) and at least one deformable element (Ref. 6, Fig. 4), where the deformable element has a first flexible arm (Ref. 8, Fig. 4-3) arranged relatively towards the cleaning zone (4) and a second flexible arm (Ref. 9, Fig. 4-3) arranged relatively away from the cleaning zone (Fig. 4), where the deformable element can move from a resting position (Fig. 4) in which the first arm (8) is substantially flat (Fig. 4) and aligned to the handle (Fig. 4), to a cleaning position (Fig. 3) in which the first arm (8) and the cleaning zone (4) adopt a curvature of a surface of a dentition (Fig. 3), where the arms form a wedge (Fig. 4) being connected together at one end of the deformable element (Fig. 4 show the arms (8&9) are connected on one end (right side)) and distanced from each other at the other end of the deformable element (Fig. 4 show the arms (8&9) are connected on one end (left side)), where the first and second arms are linked by at least two bridges (Ref. 10, Fig. 4, [0086]) which are tiltable with respect to the first and second arms when pressure is applied to the cleaning zone (fig. 3, [0097]), wherein the arms and bridges enclose a void space (Ref. 19, Fig. 1,2&4 show void spaces between the bridges (22), [0098]).

- Regarding Claim 16, Kraemer teaches the limitations of claim 15, as described above, and further teaches wherein the arms and the bridges are formed integrally by injection moulding ([0044&0111] describes elastic elements can be made by injection molding).

- Regarding Claim 17, Kraemer teaches the limitations of claim 15, as described above, and further teaches wherein the arms and the bridges are made of a thermoplastic elastomer (TPE) material ([0025,0027,&0081] describes the arm made of a thermoplastic elastomer).

- Regarding Claim 19, Kraemer teaches the limitations of claim 15, as described above, and further teaches wherein the bridges have hinges (Ref. 15&17, Fig. 2A) flexibly connecting the bridges to the arms (Fig. 2A-3, [0087]).

- Regarding Claim 20, Kraemer teaches the limitations of claim 19, as described above, and further teaches wherein the hinges are integral hinges (Fig. 2A).

- Regarding Claim 24, Kraemer teaches the limitations of claim 15, as described above, and further teaches wherein the deformable element is not longer than 150% of the length of the cleaning zone (Fig. 4 shows the length is not longer than 100% of the cleaning zone), preferably not longer than 130% of the length of the cleaning zone (Fig. 4 shows the length is not longer than 100% of the cleaning zone).

- Regarding Claim 25, Kraemer teaches the limitations of claim 15, as described above, and further teaches wherein the deformable element has 2-7 bridges, preferably 3-5 bridges, most preferably 3 bridges (Fig. 2).

- Regarding Claim 26, Kraemer teaches the limitations of claim 15, as described above, and further teaches wherein the handle comprises a polypropylene skeleton and is formed integrally in one piece with the deformable element by injection moulding ([0050-0052 & 0081-0082] describes the handle and wing are integrally made of polypropylene by injection molding forming a plastic skeleton).

- Regarding Claim 28, Kraemer teaches the limitations of claim 15, as described above, and further teaches wherein an angle (a) between the bridges and the arms is around 90° in the resting position (Fig. 4), and around 95° to 110° preferably around 97° to 105° when pressure is applied to the cleaning zone (Fig. 3).

Claim Rejections - 35 USC § 103

8. The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

9. **Claims 18 and 21-23** are rejected under 35 U.S.C. 103 as being unpatentable over Kraemer (2009/0313779).

- Regarding Claim 18, Kraemer teaches the limitations of claim 15, as described above, and further teaches wherein the TPE material has some Shore D hardness ([0027]). Kraemer fails to explicitly teach a Shore D hardness of about 50 to 66. Kraemer discloses that the shore D hardness amount allows the bridges to be bent (Fig. 3) and needs to be optimized to "conform to users teeth." As seen in Fig. 3, the bridges bend and move to conform to users teeth and the device is disclosed to be a result effective variable in that changing shore D hardness affects guidance and ease of conformity to teeth. Further, it appears that one of ordinary skill in the art would have had a reasonable expectation of success in modifying the Kraemer reference to have a shore D hardness within the claimed range, as it involves only adjusting the material property of a component disclosed. Therefore, it would have been obvious to one having ordinary skill in

the art before the effective filing date of the claimed invention to modify the device of Kraemer by making the shore D hardness 50 to 66 as a matter of routine optimization since it has been held that "where the general conditions of a claim are disclosed in the prior art, it is not inventive to discover the optimum or workable ranges by routine experimentation." *In re Aller*, 220 F.2d 454, 456, 105 USPQ 233, 235 (CCPA 1955). Further changing Kraemer's shore D hardness produces the same predictable result of allowing the bridges to bend to conform to teeth as required in the instant application.

- Regarding Claim 21, Kraemer teaches the limitations of claim 15, as described above, and further teaches wherein the integral hinges have a thickness in the longitudinal direction of the toothbrush (Fig. 2A, [0087] describes the hinges to be thinner than the bridges to allow for deformation). Kraemer fails to explicitly teach the thickness of the hinges to be 0.2 to 0.4mm. Kraemer discloses that the thickness of the hinges allows the bridges to be bent (Figs. 2A&3,[0087]) and needs to be optimized to "conform to users teeth." As seen in Fig. 2A&3, the thickness of the hinges allows the bridges to bend to conform to users teeth and the device is disclosed to be a result effective variable in that the thickness affects guidance and ease of conformity to teeth. Further, it appears that one of ordinary skill in the art would have had a reasonable expectation of success in modifying the Kraemer reference to have a thickness within the claimed range, as it involves only adjusting the dimension of a component disclosed to require adjustment. Therefore, it would have been obvious to one having ordinary skill in the art before the effective filing date of the claimed invention to modify the device of Kraemer by making the thickness of the hinges to be 0.2mm to 0.4mm as a matter of routine optimization since it has been held that "where the general conditions of a claim are disclosed in the prior art, it is not inventive to discover the optimum or workable ranges by routine experimentation." *In re Aller*, 220 F.2d 454, 456, 105 USPQ 233, 235 (CCPA 1955). Further changing the thickness

of the hinges in Kraemer produces the same predictable result of allowing the bridges to bend to conform to teeth as required in the instant application.

- Regarding Claim 22, Kraemer teaches the limitations of claim 15, as described above, and further teaches wherein a distance between the arms (Fig. 4 shows distance between the arms (8&9)) at a proximal end of the deformable element (Fig. 4 left and right sides without hashing). Kraemer fails to explicitly teach the distance to be between 3mm to 5mm. Kraemer discloses that the distance between the arms amount allows the bridges to be placed between and bent (Fig. 3) and needs to be optimized to "conform to users teeth." As seen in Fig. 3, the distance between the arms to conform to users teeth and the device is disclosed to be a result effective variable in that the distance affects guidance and ease of conformity to teeth. Further, it appears that one of ordinary skill in the art would have had a reasonable expectation of success in modifying the Kraemer reference to have a distance within the claimed range, as it involves only adjusting the dimension of a component disclosed to require adjustment. Therefore, it would have been obvious to one having ordinary skill in the art before the effective filing date of the claimed invention to modify the device of Kraemer by making the distance of between the arms to be 3mm to 5mm as a matter of routine optimization since it has been held that "where the general conditions of a claim are disclosed in the prior art, it is not inventive to discover the optimum or workable ranges by routine experimentation." *In re Aller*, 220 F.2d 454, 456, 105 USPQ 233, 235 (CCPA 1955). Further changing the distance in Kraemer produces the same predictable result of allowing the bridges to bend to conform to teeth as required in the instant application.

- Regarding Claim 23, Kraemer teaches the limitations of claim 15, as described above, and further teaches wherein there is a longitudinal distance between adjacent bridges (Fig. 4 shows

distance between the bridges). Kraemer fails to explicitly teach the distance to be between 5mm to 9mm. Kraemer discloses that the distance between the bridges allows the deformable region to be bent (Fig. 3) and needs to be optimized to "conform to users teeth." As seen in Fig. 3, the distance between the bridges to conform to users teeth and the device is disclosed to be a result effective variable in that the distance affects guidance and ease of conformity to teeth. Further, it appears that one of ordinary skill in the art would have had a reasonable expectation of success in modifying the Kraemer reference to have a distance within the claimed range, as it involves only adjusting the dimension of a component disclosed to require adjustment. Therefore, it would have been obvious to one having ordinary skill in the art before the effective filing date of the claimed invention to modify the device of Kraemer by making the distance of between adjacent bridges to be 5mm to 9mm as a matter of routine optimization since it has been held that "where the general conditions of a claim are disclosed in the prior art, it is not inventive to discover the optimum or workable ranges by routine experimentation." *In re Aller*, 220 F.2d 454, 456, 105 USPQ 233, 235 (CCPA 1955). Further changing the distance in Kraemer produces the same predictable result of allowing the bridges to bend to conform to teeth as required in the instant application.

10. **Claim 27** is rejected under 35 U.S.C. 103 as being unpatentable over Kraemer as applied to claim 15 above, and further in view of Precision Punch and Plastics (NPL, https://precisionpunch.com/plastic-materials-index/polypropylene/).

- Regarding Claim 27, Kraemer teaches the limitations of claim 15, as described above, and further teaches the deformable element is made of polypropylene ([0050-0052&0081]). Kraemer fails to explicitly teach wherein the deformable element consists of a material having an elongation at break of 500 to 800%. Precision and Punch teaches polypropylene having an

elongation at break of 100 to 600%. Therefore, It would have been obvious to one having ordinary skill in the art before the effective filing date of the claimed invention to modify the deformable element made of polypropylene having an elongation break from 500 to 800% since it has been held that "[i]n the case where the claimed ranges 'overlap or lie inside ranges disclosed by the prior art' a prima facie case of obviousness exists." *In re Wertheim*, 541 F.2d 257, 191 USPQ 90 (CCPA 1976); *In re Woodruff*, 919 F.2d 1575, 16 USPQ2d 1934 (Fed. Cir. 1990). Further, since the polypropylene is known to have an elongation break that overlaps with the claimed range, it would have been obvious to one of ordinary skill in the art before the effective filing date to modify the elongation at bread to be between 500 to 800% since such a modification would yield the predictable result of allowing the deformable element to conform to teeth.

Conclusion

11. The prior art made of record and not relied upon is considered pertinent to applicant's disclosure. Kramer (6,327,735), Berl (4,694,844), Blakeman (2,706,825), and Chodorow (WO 0121035 A1) teaches toothbrushes with a deformable element and can be considered analogous art because they are within the same field of endeavor.

12. Any inquiry concerning this communication or earlier communications from the examiner should be directed to DANA L POON whose telephone number is (571)272-6164. The examiner can normally be reached on General: 6:30AM-3:30PM.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice. If attempts to reach the examiner by telephone are unsuccessful, the examiner' s supervisor, David Posigian can be reached on (313) 446-6546. The fax phone number for the organization where this application or

proceeding is assigned is 571-273-8300. Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see https://ppairmy.uspto.gov/pair/PrivatePair. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/DANA LEE POON/
Examiner, Art Unit 3723